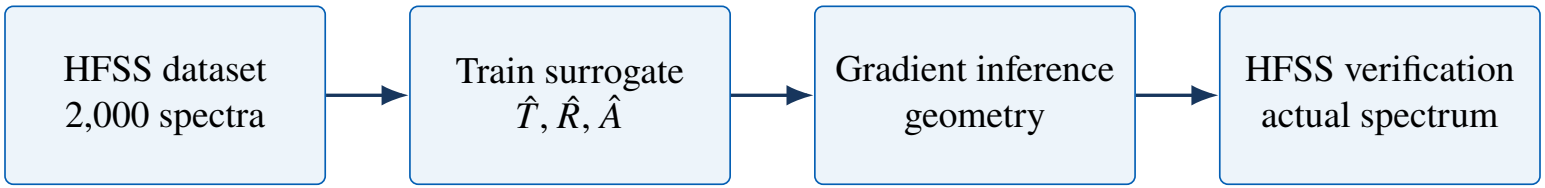
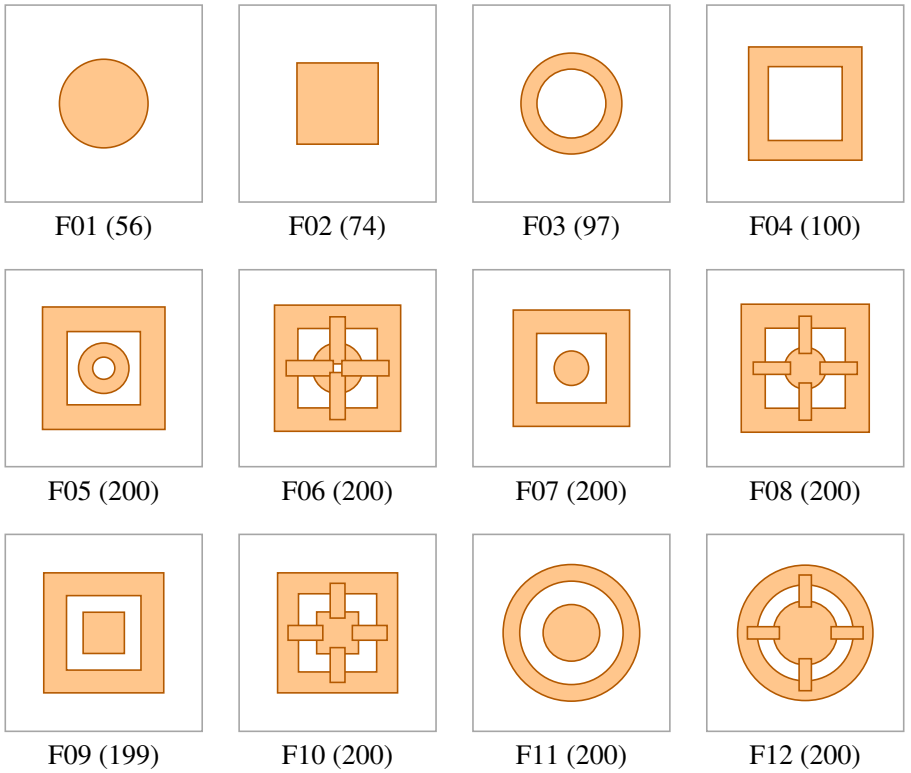
Problem and workflow



- Request: target absorption over a wavelength band.
- A trained surrogate guides gradient-based geometry search.
- HFSS supplies the physical check.

Takeaway: The workflow connects received HFSS data to an auditable physical verification.

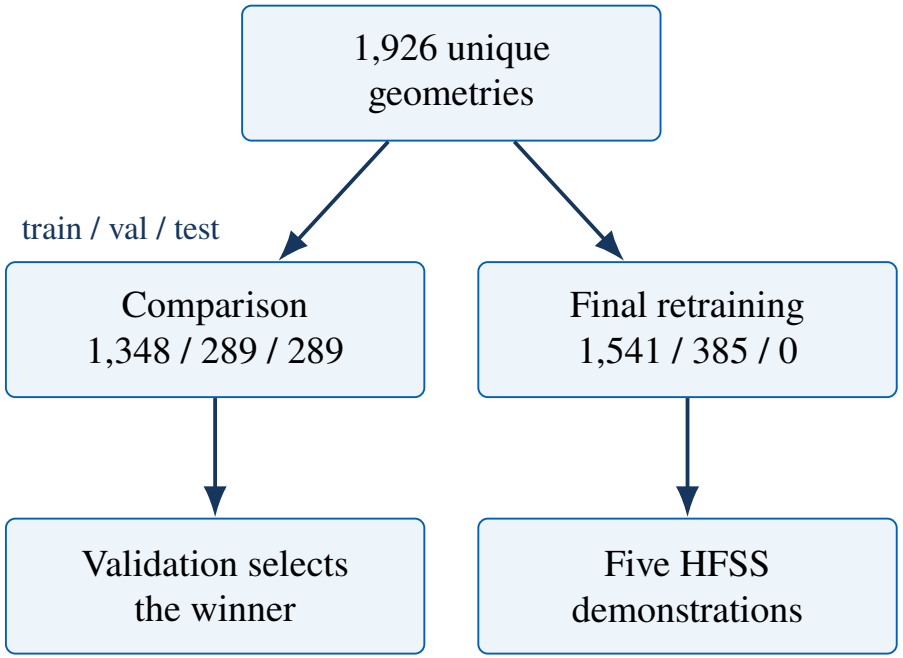
Received air-only dataset



Twelve received geometry families. Counts are unique geometries after duplicate grouping.

Takeaway: Keeping duplicate geometries together prevents an inflated test metric.

Two uses of the same dataset



Model input

- family ID
- seven native parameters
- family mask

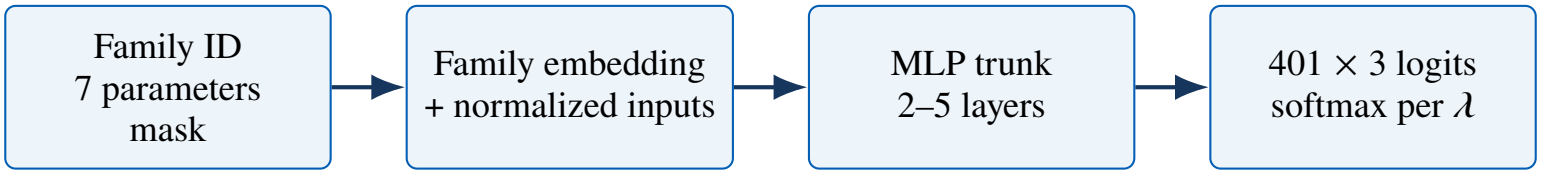
Model output

- 401 wavelengths
- T, R, A with a softmax at each wavelength

Takeaway: Held-out metrics and physical HFSS results belong to different model sets.

Symbols: T, R, A: transmitted, reflected, and absorbed power fractions. Splits are train / validation / test geometry counts.

Surrogate model and training objective



Loss: absorption MSE + maximum spectral error
+ auxiliary $T/R/A$ MSE

Training: 300 epoch cap, patience 30, batch size 64. Atomic model, optimizer, and random-state recovery enables resumption.

Takeaway: The loss targets absorption directly, including its worst-case error per spectrum, with $T/R/A$ as an auxiliary term.

Symbols: MLP: multilayer perceptron. λ: wavelength. The softmax enforces $T + R + A = 1$ but does not prove physical realizability.

Four independent comparison cases

Case	selection	selected topology
Baseline	fixed	residual, 256×3, dropout 0.1
Optuna	100 candidates	plain, 256×3, dropout 0
Ray	100 early-stopped candidates	plain, 256×3, dropout 0
AutoKeras	AutoModel then PyTorch export	plain, 128×3

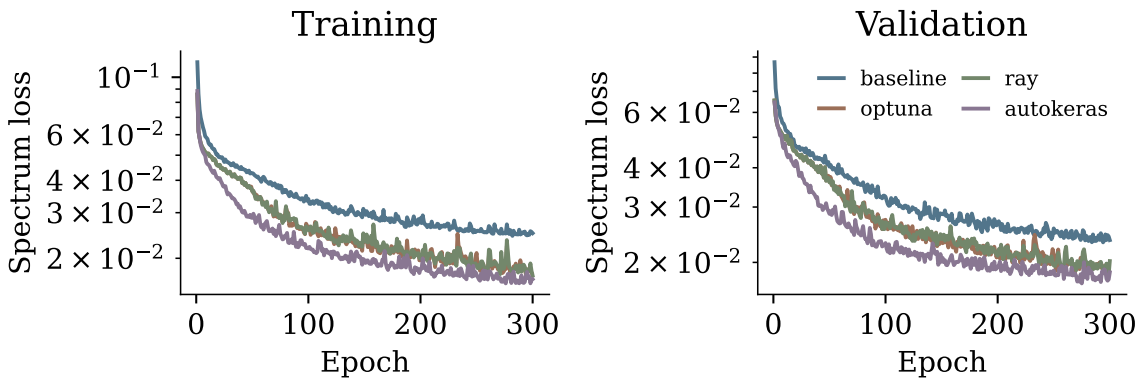
Search space

Plain or residual trunk, width 128/256/512, depth 2–5, dropout 0/0.1/0.2.

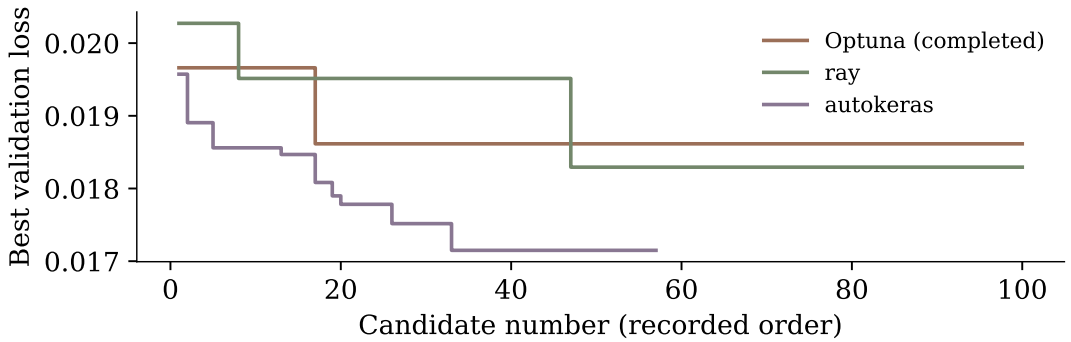
Optuna: 54 completed and 46 pruned candidates. AutoKeras: 57 completed.

Takeaway: Optuna and Ray selected the same topology, but remain independent comparison cases.

Training and search results



Training and validation histories.

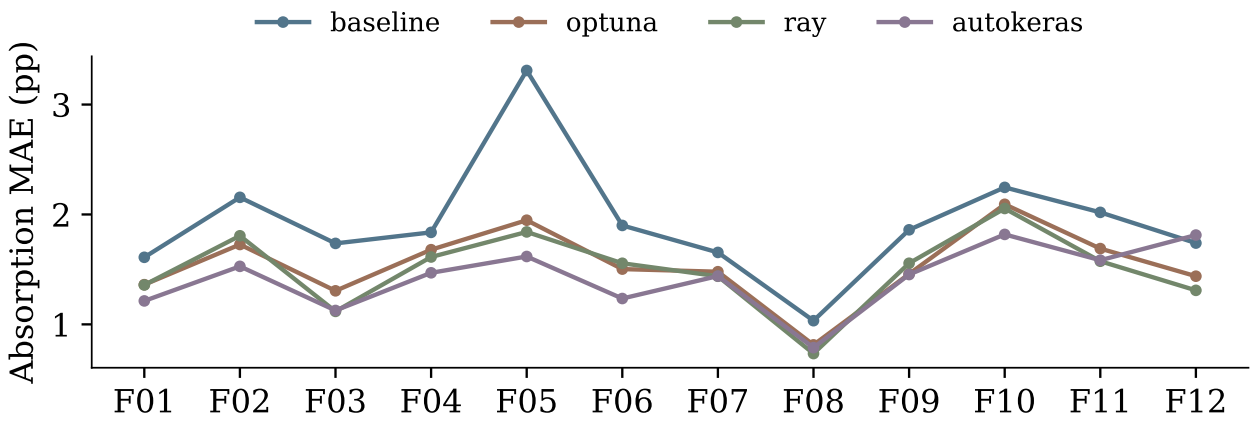


Best validation loss by search trial.

Takeaway: The searched cases improve validation loss relative to the fixed baseline.

Error on 289 held-out geometries

Case	MAE (pp)	mean max (pp)	P95 max (pp)
AutoKeras	1.45	6.52	13.02
Ray	1.50	6.87	13.28
Optuna	1.55	6.92	12.95
Baseline	1.95	8.48	15.90

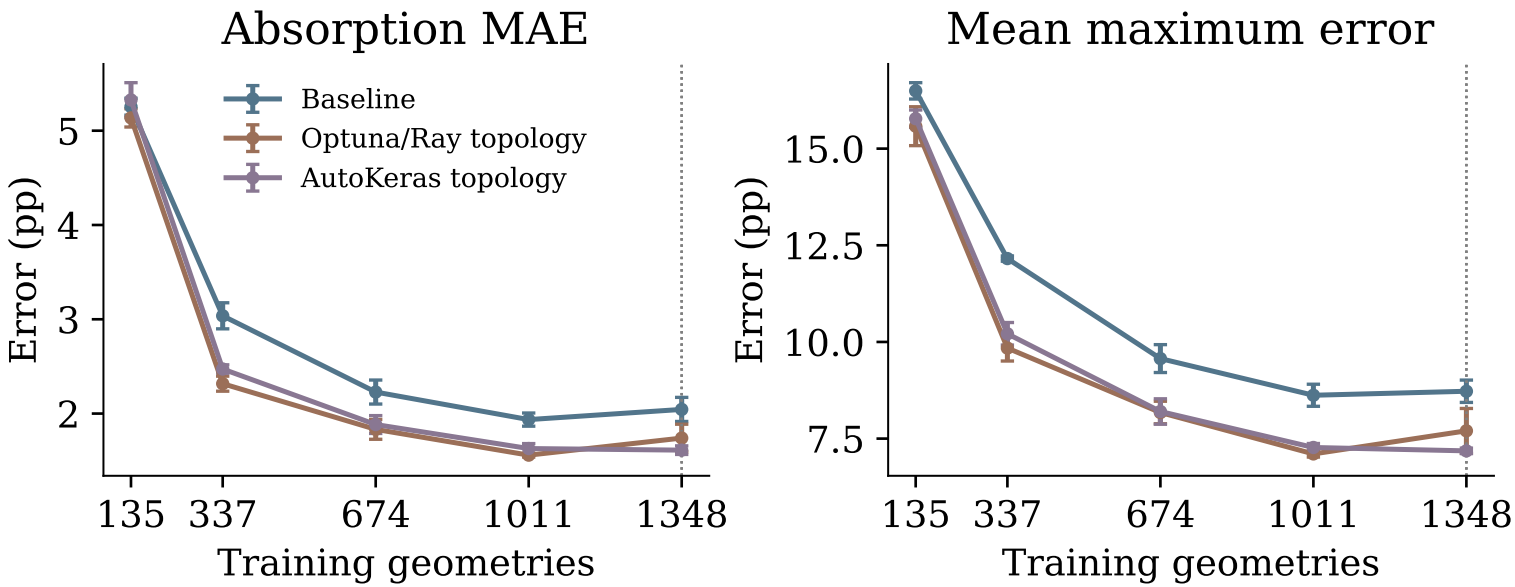


Absorption MAE per geometry family.

Takeaway: Absorption MAE spans 1.45–1.95 percentage points across the four cases.

Symbols: MAE: mean absolute absorption error. mean max and P95 max: mean and 95th percentile of the largest absorption error per spectrum. pp: percentage points.

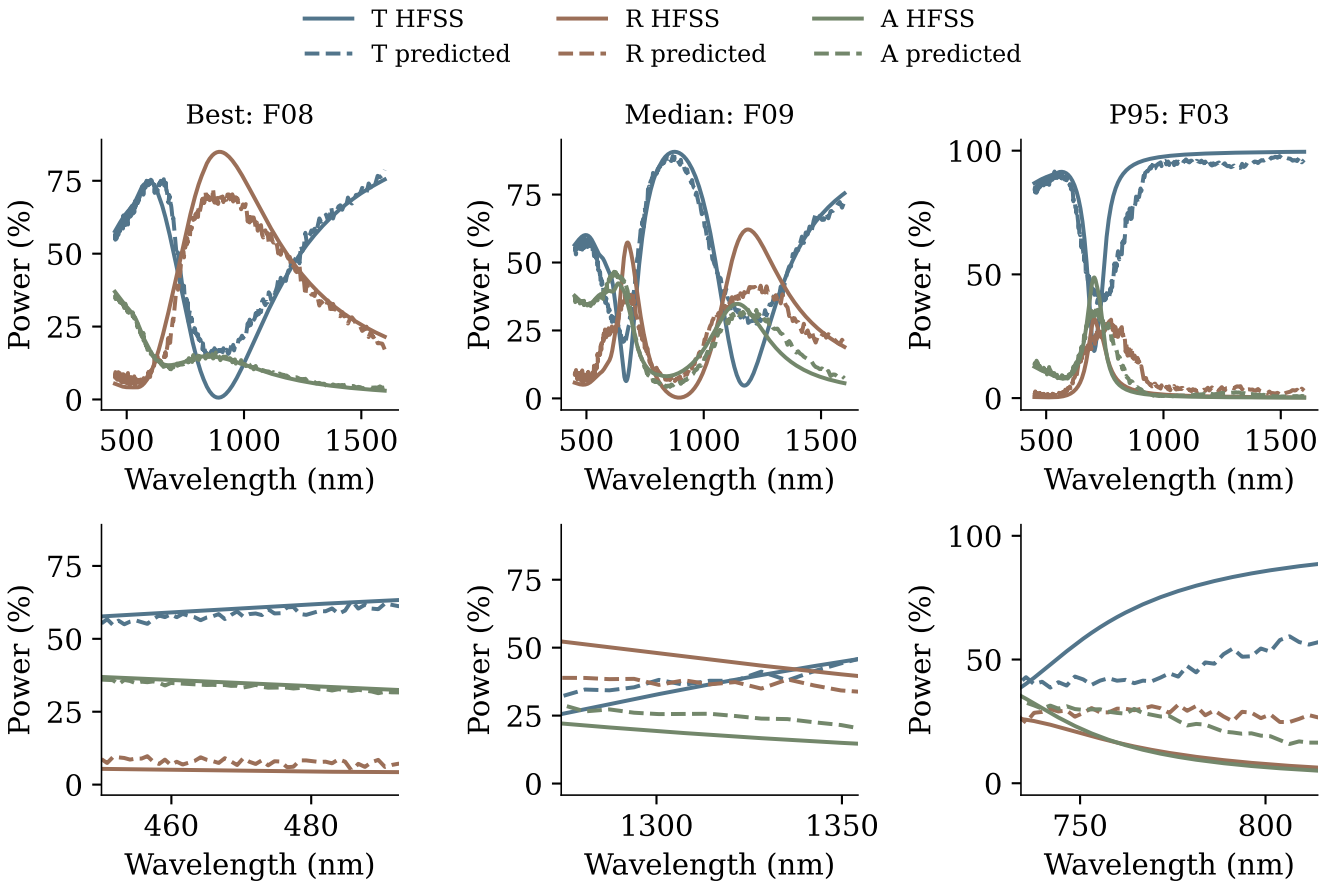
Effect of training-set size



Nested family-stratified subsets of the fixed 1,348 training geometries, three seeds per point. Error bars: sample SD over three seeds. Topologies selected with 1,348 rows are held fixed.

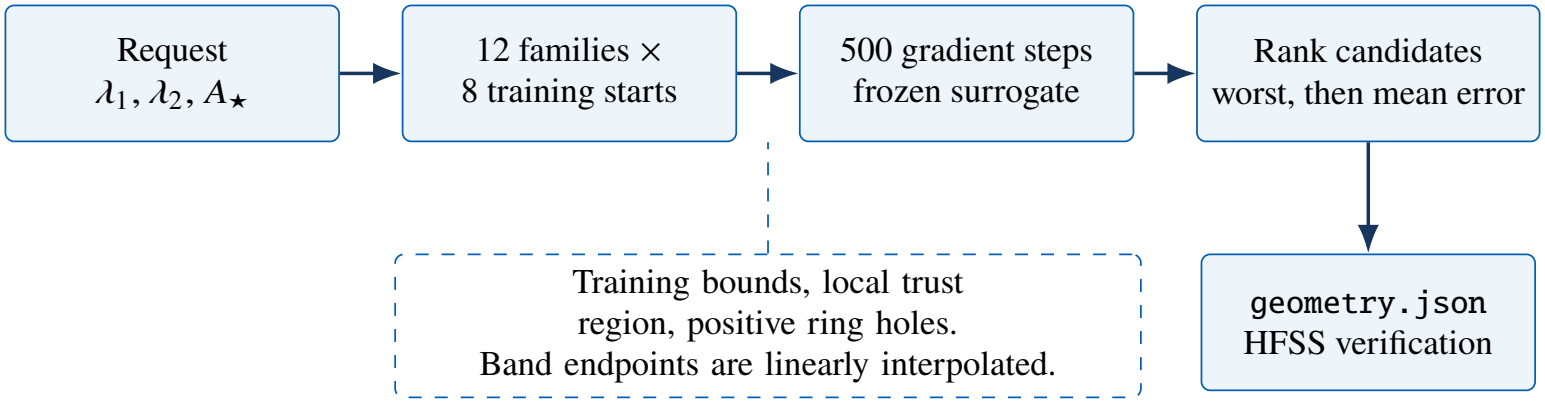
Takeaway: Error drops strongly up to 674–1,011 geometries. The last step is not uniformly better. Silica data remain pending.

Held-out spectra selected by a fixed rule



Takeaway: Best, median, and P95 examples are fixed quantiles of baseline maximum absorption error, not hand-picked spectra.

Inference under geometry constraints

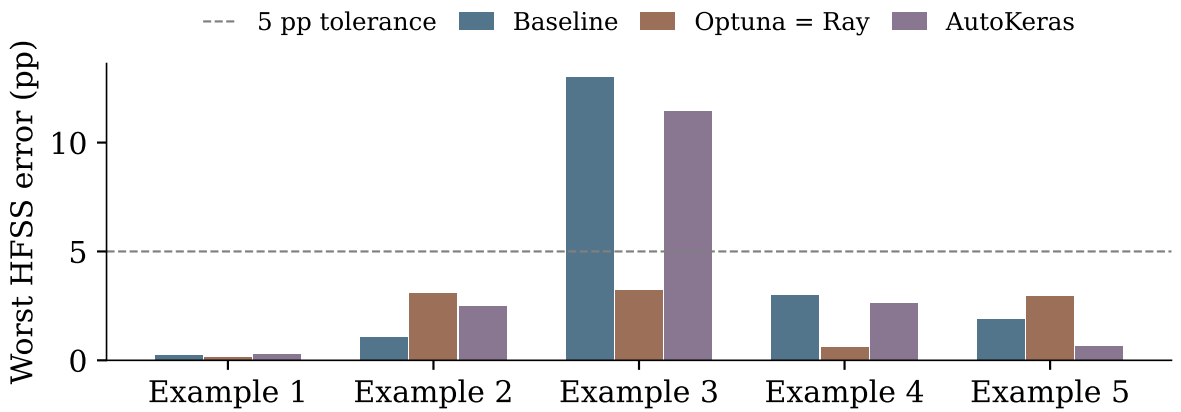


$$\min_{\theta} \max_{\lambda \in [\lambda_1, \lambda_2]} \left| \hat{A}(\lambda; \theta) - A_\star \right|, \text{ then wavelength-weighted mean error.}$$

Takeaway: Candidates remain inside training bounds and local trust regions, with positive ring holes.

Symbols: λ_1, λ_2: band edges. $A_\star$: target absorption. $\hat{A}$: surrogate absorption. θ: native geometry parameters of one family.

Five narrow-band HFSS demonstrations

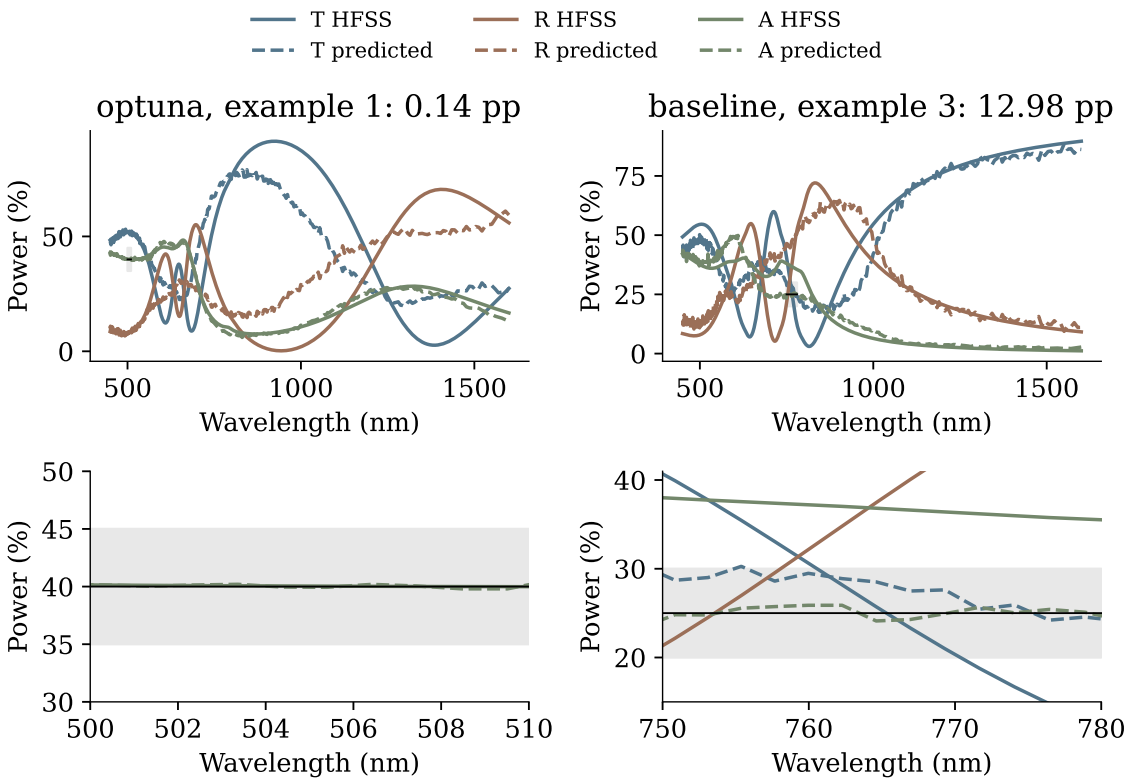


Worst HFSS error (pp)	Ex. 1 500–510 40%	Ex. 2 600–620 30%	Ex. 3 750–780 25%	Ex. 4 1000–1040 20%	Ex. 5 1250–1300 15%	Mean	≤ 5 pp
Baseline	0.23	1.05	12.98	2.99	1.90	3.83	4/5
Optuna = Ray	0.14	3.08	3.22	0.61	2.94	2.00	5/5
AutoKeras	0.30	2.49	11.44	2.61	0.63	3.49	4/5

Takeaway: These five dataset-supported demonstrations are not a blind generalization test.

Symbols: Ex.: request, with band in nm and target absorption. Error: largest in-band difference between HFSS absorption and target.

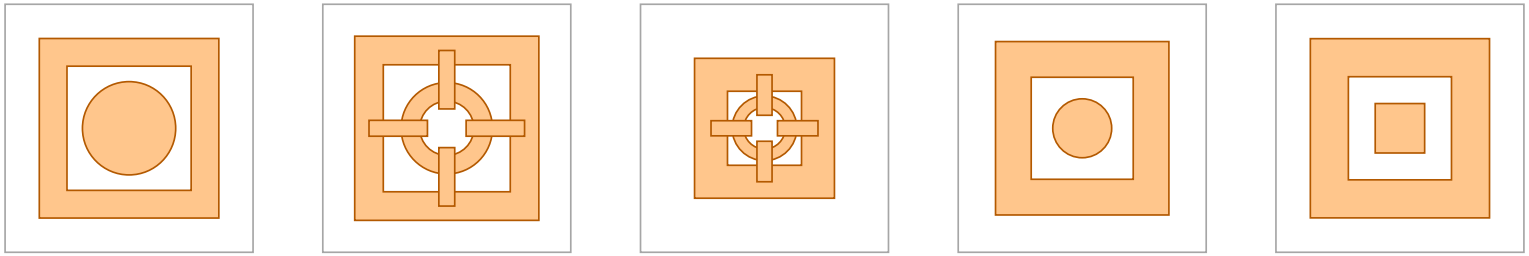
HFSS spectra include a difficult case



Baseline, example 3: surrogate worst error 0.88 pp, HFSS worst error 12.98 pp.

Takeaway: Physical verification exposes surrogate optimism even when the predicted target error is small.

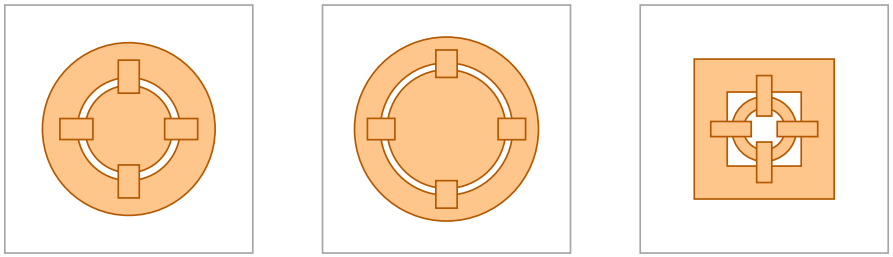
Geometries returned by inference



1: F07 2: F06 3: F06 4: F07 5: F09

Optuna winners, identical 320 nm air cells. Circular sizes are radii.

Example 3, model-specific native geometries



Baseline F12 AutoKeras F12 Optuna/Ray F06

Each geometry is also saved in native nanometres as `geometry.json`.

Takeaway: The output is a native-nanometre geometry file ready for a received-template HFSS verification.

Conclusions and next steps

Conclusions

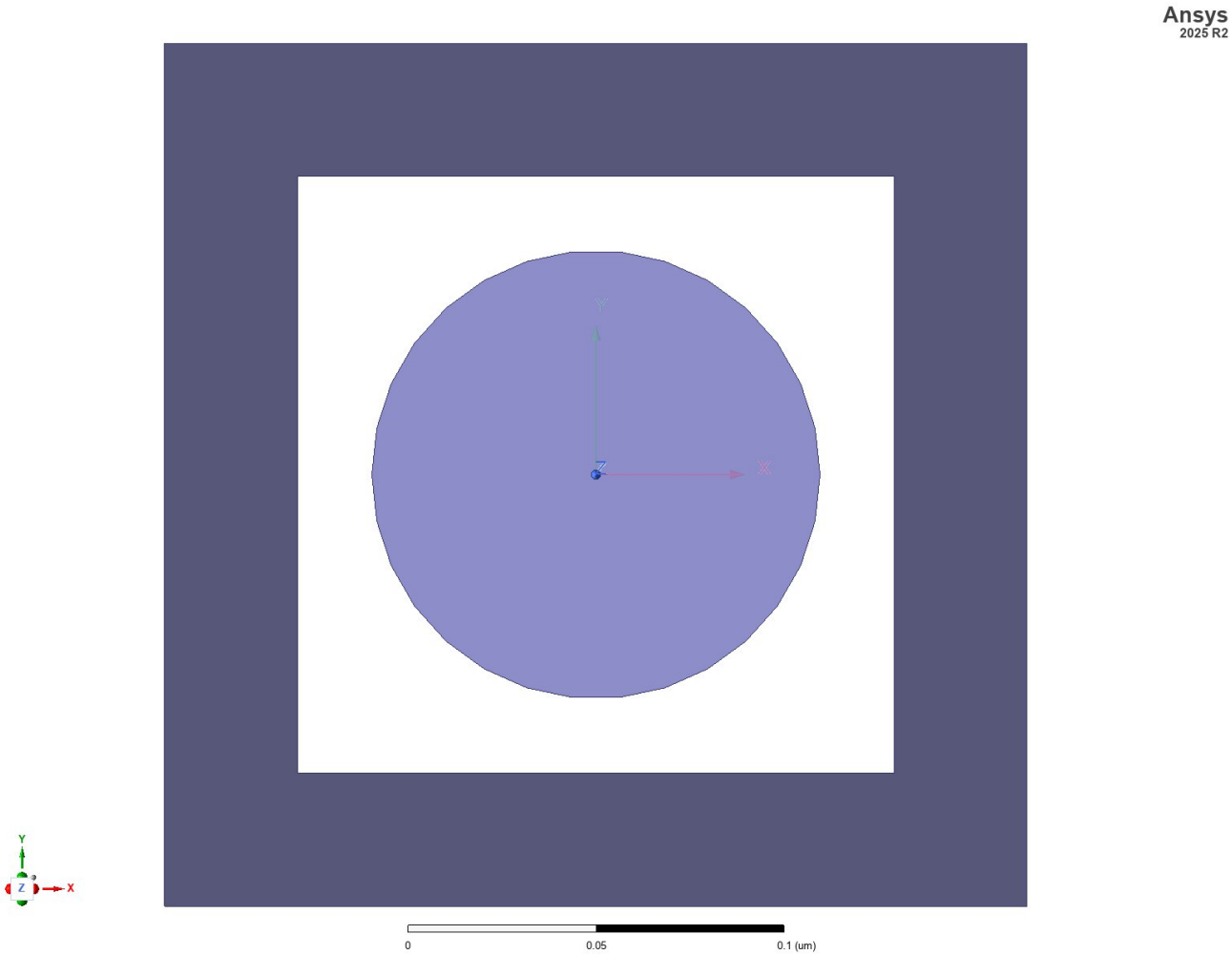
- End-to-end data, training, inference, and HFSS workflow operates.
- Held-out absorption MAE is 1.45–1.95 pp.
- Most narrow-band demonstrations meet ±5 pp in HFSS.

Next steps

- Receive and prepare 2,000 silica spectra.
- Run more HFSS validations.
- Reassess data-size learning curves.

Takeaway: The evidence currently supports air-only, narrow-band demonstrations with explicit physical verification.

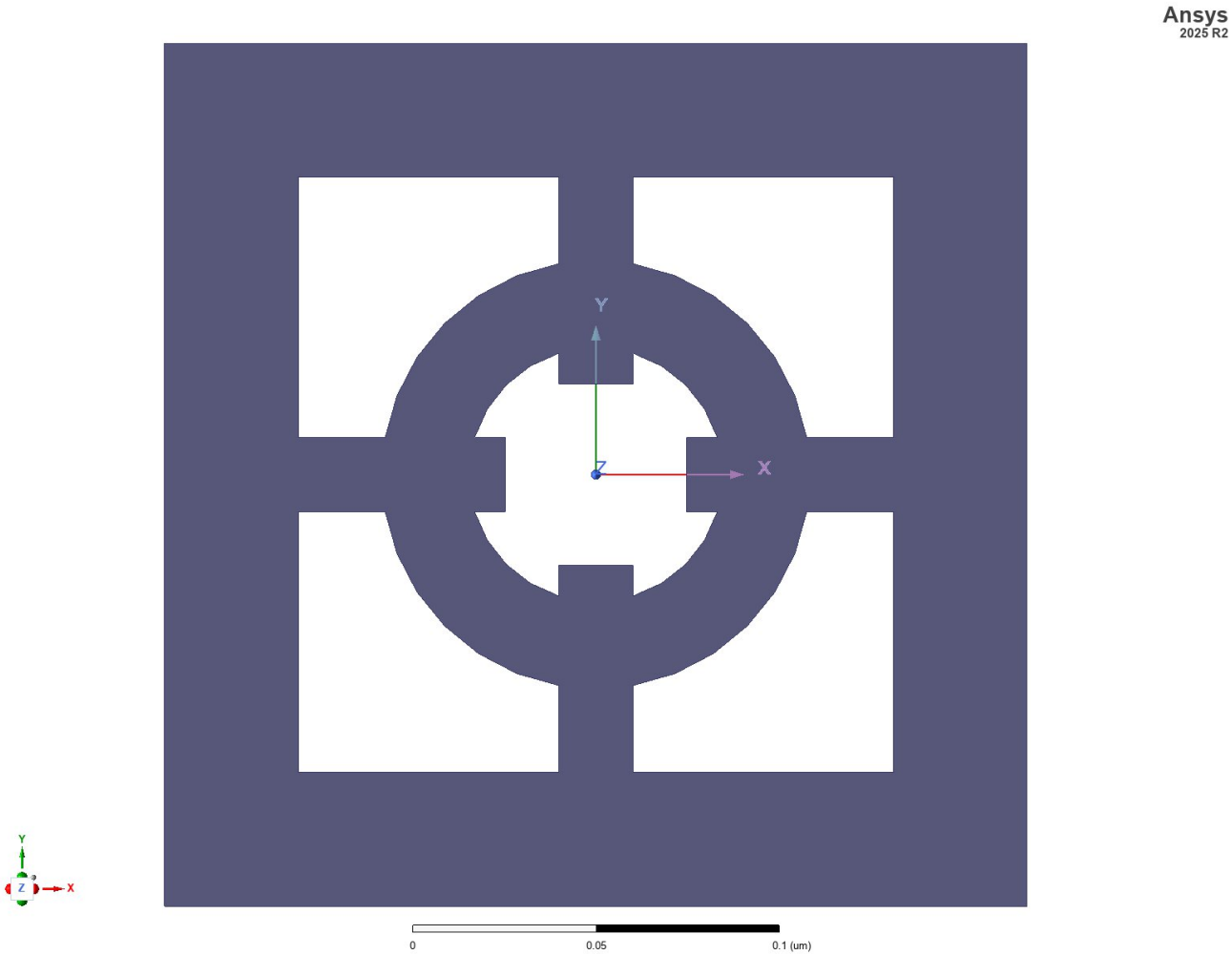
Backup: HFSS geometry, example 1



Optuna = Ray, F07. Target: 40% over 500–510 nm.

Top view exported from the verified HFSS project.

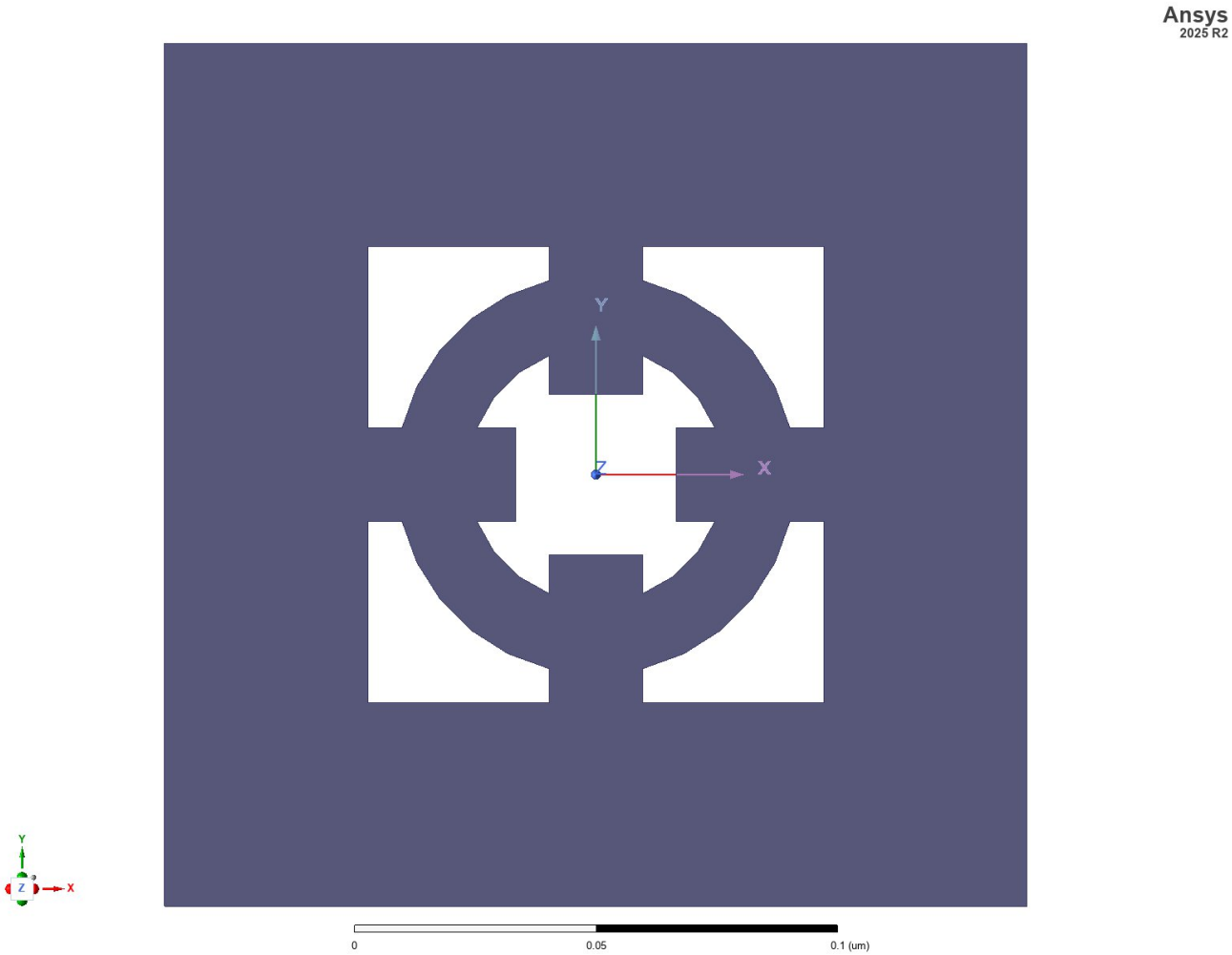
Backup: HFSS geometry, example 2



Optuna = Ray, F06. Target: 30% over 600–620 nm.

Top view exported from the verified HFSS project.

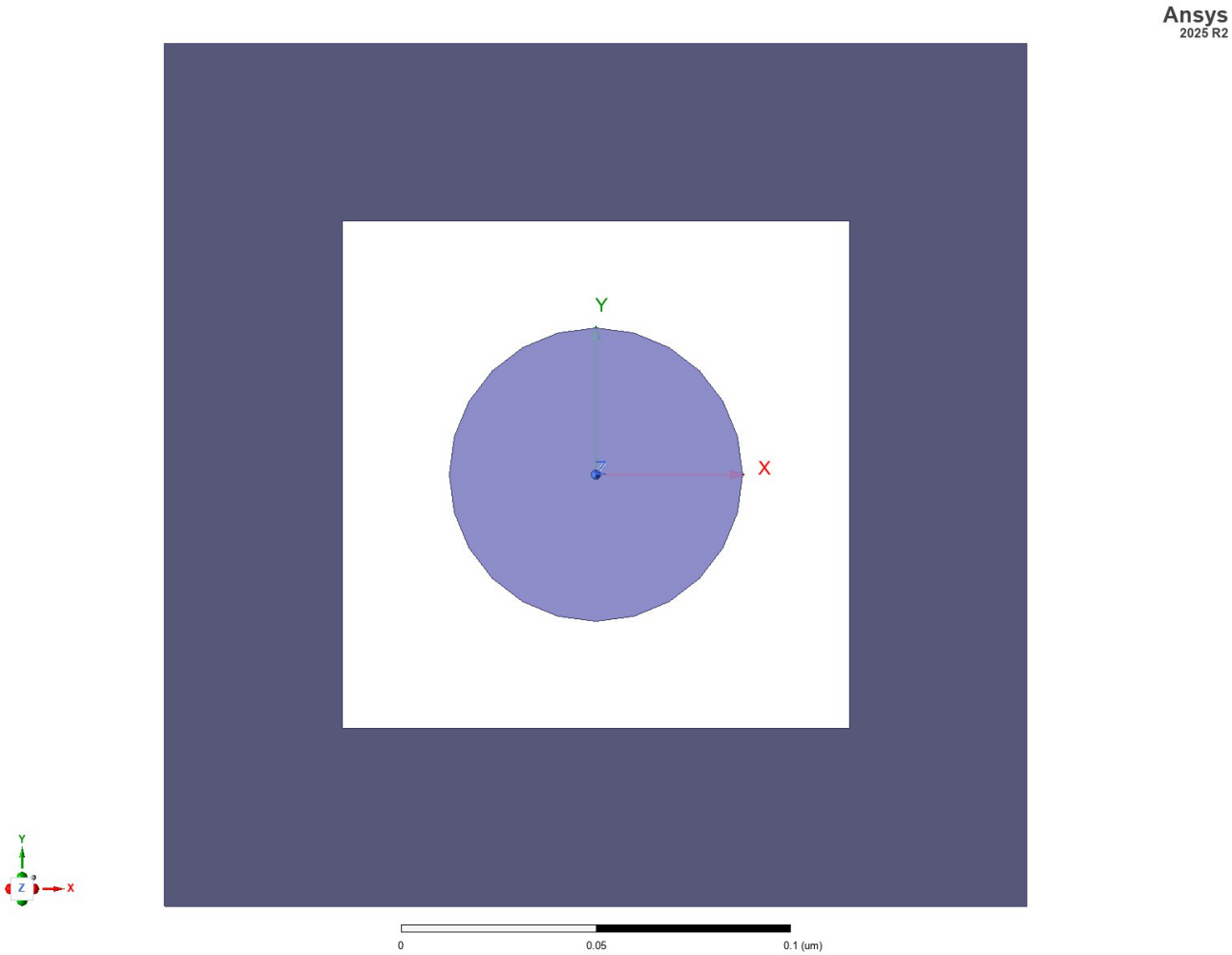
Backup: HFSS geometry, example 3



Optuna = Ray, F06. Target: 25% over 750–780 nm.

Top view exported from the verified HFSS project.

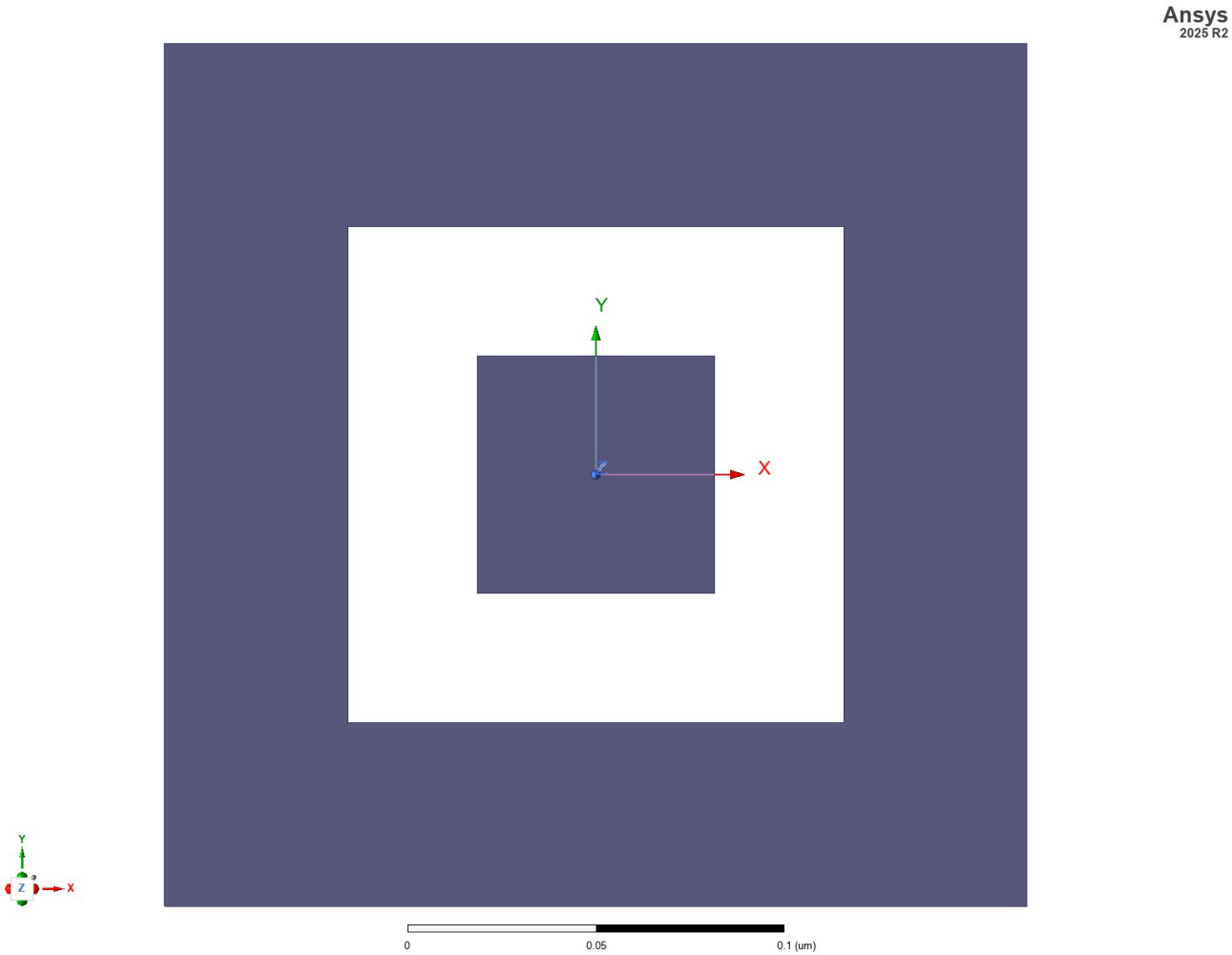
Backup: HFSS geometry, example 4



Optuna = Ray, F07. Target: 20% over 1000–1040 nm.

Top view exported from the verified HFSS project.

Backup: HFSS geometry, example 5



Optuna = Ray, F09. Target: 15% over 1250–1300 nm.

Top view exported from the verified HFSS project.